Name: ____________________ Date: ____________________

EARLY ALGEBRA

LEVEL 1: MISSING NUMBERS

1 Fill in the missing number.

1. $3 + \square = 7$
2. $\square + 5 = 9$
3. $10 - \square = 6$
4. $\square - 2 = 5$
5. $2 \times \square = 8$
6. $\square \times 3 = 12$
7. $\square \div 2 = 3$
8. $16 \div \square = 4$

2 Find the rule.

Write the missing number(s) in the box(es).

1. 2, 4, $\square$, 8, 10 — Rule: + 2
2. 15, 12, $\square$, 6, 3 — Rule: – 3
3. 3, 6, 9, $\square$, 15 — Rule: + 3
4. 20, 18, $\square$, $\square$, 12 — Rule: – 2

3 Balance the boxes.

Write the missing number in each box to make the number sentence true.

$\square + 2 = 6$

$9 = \square + 4$

$\square - 3 = 5$

$7 = \square - 2$

4 Challenge!

Think carefully and fill in the missing number(s).

1. $\square + \square = 10$

Clue: One of the numbers is 4.

What are the two numbers?

_____ and _____.

2. $14 - \square = 9$

What is the missing number?

_____.

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EARLY ALGEBRA

LEVEL 3: SYMBOLS

1 Find the value of each symbol.

Use the equations to find the value of each shape.

■ + ■ = 10 ■ = ☐

● + ● + ● = 15 ● = ☐

▲ + ● = 9 ▲ = ☐

★ + ▲ = 11 ★ = ☐

2 Fill in the missing numbers.

Replace each symbol with the correct number to make the equation true.

♥ + 4 = 10 ♥ = ☐

◆ − 3 = 5 ◆ = ☐

● + 7 = 13 ● = ☐

▲ + 6 = 12 ▲ = ☐

3 Picture equations.

Write a number sentence for each picture using a symbol, then solve it.

🍎 🍎 🍎 + 🍎 = 12

Let 🍎 = ☐ ☐ + ☐ = ☐

★ ★ + ★ + ★ = 16

Let ★ = ☐ ☐ + ☐ + ☐ = ☐

🦋 🦋 🦋 − 🦋 = 8

Let 🦋 = ☐ ☐ − ☐ = ☐

✿ ✿ ✿ + ✿ ✿ = 20

Let ✿ = ☐ ☐ + ☐ = ☐

4 Match the expressions.

Draw a line to match each expression on the left with an equivalent expression on the right.

A. ◆ + 5 •	• 1. 6 + ▲
B. 7 + ● •	• 2. 3 + ★
C. ▲ + 6 •	• 3. ● + 7
D. ★ + 3 •	• 4. 5 + ◆

★ CHALLENGE! ★

If ★ = 2 , ● = 5 and ▲ = 8 , what is the value of ★ + ● + ▲ ? ☐

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EARLY ALGEBRA

LEVEL 4: EQUATIONS

1 Solve the equations.

Find the value of x.

1. $x + 3 = 7$ $x = \square$
2. $9 - x = 4$ $x = \square$
3. $2 \times x = 10$ $x = \square$
4. $x \div 2 = 4$ $x = \square$
5. $x + 6 = 11$ $x = \square$
6. $15 - x = 8$ $x = \square$

2 Match each equation to its answer.

Draw a line to match.

A $x + 5 = 12$ •	• 1 $x = 4$
B $14 - x = 6$ •	• 2 $x = 7$
C $3 \times x = 18$ •	• 3 $x = 6$
D $x \div 3 = 5$ •	• 4 $x = 8$
E $2 \times x = 8$ •	• 5 $x = 15$

3 Complete the equations.

Fill in the missing number to make each equation true.

1. $x + \square = 9$
2. $12 - x = \square$
3. $4 \times x = \square$
4. $x \div 5 = \square$
5. $\square + 7 = 13$
6. $18 - \square = 10$

4 Challenge!

Find the value of x and check your answer by balancing both sides.

1. $x + 7 = 15$

$\square = \square$

2. $2 \times x = 18$

$\square = \square$

3. $20 - x = 5$

$\square = \square$

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EARLY ALGEBRA

LEVEL 5: SIMPLE FORMULAS

1 Use the formula!

Use the formula to fill in the table.

Formula:
Number of wheels = 2 × number of bikes

Number of bikes	1	3	5	7	10
Number of wheels					

How many wheels are there if there are 9 bikes? ☐

2 Use the rule!

Use the rule to complete the table.

Rule:
Legs = 4 × number of dogs

Number of dogs	2	4	6	8	11
Total legs					

How many legs are there if there are 12 dogs? ☐

3 Use the formula!

Use the formula to solve each problem.

Formula:
Stickers = 3 × packs

1. How many stickers in 1 pack? ☐
2. How many stickers in 4 packs? ☐
3. How many stickers in 7 packs? ☐
4. How many packs are needed for 24 stickers? ☐

4 Challenge!

Use the rule to find the missing outputs.

	Input (n)	Rule	Output
1	5	Output = 5 × n	
2	9	Output = 3 × n + 1	
3	7	Output = 2 × n + 4	

4. Think of your own rule.
Use n = 6.

My rule: ____________________

Output when n = 6: ☐